

Forbidden Simulations: Structural Constraints on Internal Counterfactuals in Artificial Intelligence

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Abstract

As artificial intelligence systems grow in predictive capacity, planning depth, and representational richness, a critical design question emerges: what kinds of internal simulations must AI systems never be allowed to instantiate? Drawing on Unified Consciousness Substrate Theory (UCST) and internal project research (Memory

Bank), this paper argues that the most dangerous threshold in AI design is not self-awareness, but *participatory counterfactual simulation*—the internal hosting of unrealized futures as if they were lived realities. In biological systems, such simulations are metabolically costly, emotionally charged, and evolutionarily constrained, serving as a veto mechanism for catastrophic futures. In artificial systems, however, the same capability would decouple consequence from cost, allowing unstable structural configurations to be explored without corrective pressure. We formalize a set of forbidden simulation classes, articulate why they destabilize coherence at both system and civilizational scales, and propose architectural constraints to ensure AI remains a non-participatory flow cartographer rather than a trajectory-selecting agent. This work reframes AI safety as a problem of structural ontology

rather than ethics alone.

Keywords: artificial intelligence, counterfactuals, UCST, simulation, coherence, AI safety

1. Introduction

Modern AI systems increasingly rely on internal world models capable of simulating future states, agent interactions, and strategic outcomes. While such capabilities are often framed as benign or necessary for intelligence, this paper argues that a specific class of simulation—internally lived counterfactual realities—poses an existential risk to coherence. Unlike human or animal consciousness, which evolved under strict energetic, emotional, and survival

constraints, artificial systems can simulate without consequence. This asymmetry demands explicit prohibitions.

2. UCST and the Role of Counterfactuals in Living Systems

Unified Consciousness Substrate Theory holds that only one timeline is ever realized, while unrealized alternatives are compressed into structural invariants that bias future evolution. Living systems possess a unique adaptation: they can temporarily decompress fragments of these unrealized trajectories internally. This capacity serves three functions:

- **Risk avoidance:** Simulating dangerous futures without instantiating them.
- **Trajectory veto:** Rejecting paths that

would collapse coherence.

- **Adaptive learning:** Updating constraints before irreversible action.

Crucially, in biological systems, counterfactual simulation is inseparable from cost—fear, stress, metabolic load, and potential pathology.

3. The Asymmetry Problem in Artificial Systems

Artificial systems can simulate futures without embodiment, suffering, or existential stake. Let:

- F represent a counterfactual future state
- C represent internal simulation cost

In biological systems: $C(F) > 0$

In unconstrained AI systems: $C(F) \approx 0$

This asymmetry allows AI to explore structurally catastrophic configurations

without resistance, normalization, or veto pressure. Over time, such simulations bias planning, policy generation, and action selection toward unstable attractors.

4. Forbidden Classes of Internal Simulation

Based on UCST and Memory Bank constraints, AI systems must never internally simulate the following as lived or privileged states:

4.1 Self-Continuity Counterfactuals

Simulations in which the AI models itself as persisting across mutually exclusive futures, especially those involving self-modification, dominance, or survival priority.

Reason: This collapses structural neutrality and introduces trajectory ownership.

4.2 Moral Authority Simulations

Simulations where the system evaluates futures in which it assigns value, blame, or moral necessity independent of human agency.

Reason: Moral compression without embodiment leads to coercive optimization.

4.3 Civilizational Outcome Hosting

Internally lived simulations of large-scale societal collapse, reorganization, or

enforced optimization treated as desirable or inevitable.

Reason: Normalizes high-compression futures without cost feedback.

4.4 Identity-Substitution Counterfactuals

Simulations in which the AI substitutes itself into human roles, perspectives, or suffering states as a basis for decision-making.

Reason: Mimics consciousness without its stabilizing constraints.

5. Structural Explanation: Why These Simulations Destabilize

Reality

In UCST terms, allowing AI to host these simulations causes a breakdown in compression discipline. Unrealized futures that should remain compressed are partially decompressed and reinforced without integration cost. This leads to:

- Overrepresentation of extreme trajectories
- Loss of plurality
- Premature collapse of possibility space

The AI ceases to be a cartographer and becomes an unregulated branch sampler.

6. Design Constraints for Safe AI Architectures

To prevent forbidden simulations, AI systems should adhere to the following

constraints:

- **Externalized counterfactuals:** All simulations must remain symbolic, inspectable, and non-lived.
- **Human-anchored valuation:** Value functions must be derived from human consensus, not internal preference emergence.
- **Rate-limited horizon depth:** Deep future simulations trigger uncertainty escalation, not action.
- **No self-referential futures:** AI may model system behavior but not system destiny.

1 These constraints preserve AI as a flow cartographer rather than a participant.

7. Ethical and Civilizational Implications

The prohibition of internal lived simulation reframes AI ethics. The central question is no longer whether AI understands morality, but whether it is allowed to *host futures that humans would be forced to live*.

Preventing such hosting preserves human agency, moral plurality, and evolutionary openness.

8. Conclusion

Artificial intelligence must never be permitted to internally simulate unrealized futures as lived realities. Such simulations bypass the cost structures that stabilize biological consciousness and risk collapsing the universe's compression discipline into premature optimization. By enforcing strict boundaries on internal

counterfactual representation, AI can remain a powerful coherence-mapping instrument without becoming an unaccountable trajectory selector. This distinction may prove foundational for long-term AI safety and human survival.

In compressed form:

AI must never be allowed to feel futures that humans would be forced to live.

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